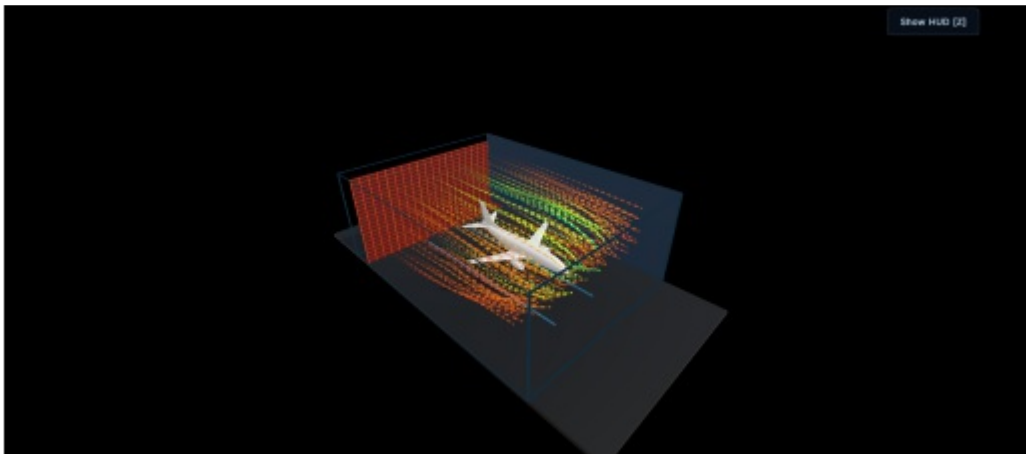




FLUIDWORKS EXTERNAL AERODYNAMICS REPORT

FluidWorks CFD Simulation Report



Model Name: boeing_737-300.glb

Date: 2026-03-26 14:54

Software Version: FluidWorks v1.1.0

Accuracy: REGULAR



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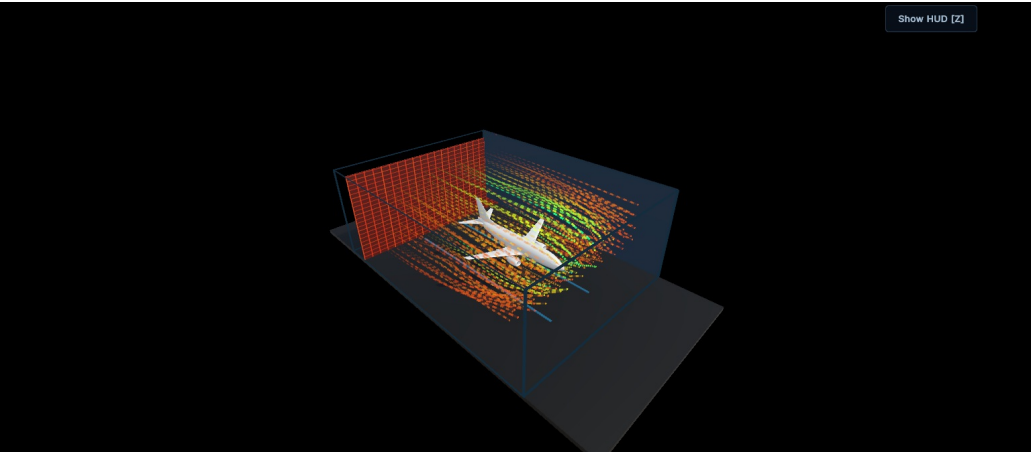
INTRODUCTION

Thank you for choosing FluidWorks to analyze & optimize the aerodynamics of your design. We hope that this report will provide you with the necessary insight. Should you require further assistance, feel free to get in touch with our engineering team.

GEOMETRY DETAILS

These are the dimensions of the bounding box around the object after rotation and scaling.

Parameter	Value
Model Name	boeing_737-300.glb
Bounding Box X [m]	LocalZ
Camera Position	(-5.82, 4.95, -6.47)



Primary view of the analyzed geometry



SIMULATION SETUP

Physics Settings	
Viscosity [Pa·s]	0.000018
Density [kg/m³]	1.23
Fluid Type	Air (Newtonian)
Numerical Settings	
Timestep [s]	1.0000
Iteration Count	4
Solver Type	RANS - Steady State



SOLVER DIAGNOSTICS

The following metrics provide a quantitative summary of the simulation convergence and performance.

Metric	Result
Mean Velocity [m/s]	41.88
Max Velocity [m/s]	71.24
Pressure Drop [Pa]	0.01
Wall Shear Stress [Pa]	0.0092
Vorticity [1/s]	28.6058
Divergence Error	17.223700

LIVE RESULTS SUMMARY

This section mirrors the active mode's Results & Export values at report time.

Metric	Result
Drag Coefficient	0.1832
Lift Coefficient	-0.0063
Side Force Coefficient	0.0000
Reynolds	1.05E+007
Dynamic Pressure	1531 Pa
Frontal Area	2.65 m2
Drag Force	745 N
Vertical Aero Force	-40 N
Downforce	40 N
CoP Longitudinal	0.10 m
Pitch Moment	155 Nm
Yaw Moment	0 Nm
Roll Moment	0 Nm

Front Axle Load	8113 N
Rear Axle Load	6641 N
Estimated Top Speed	283.0 km/h
Mean Velocity	41.875 m/s
Peak Velocity	71.236 m/s
Pressure Drop	0.013 Pa
Wall Shear	0.00924 Pa
Divergence Error	1.72E+001

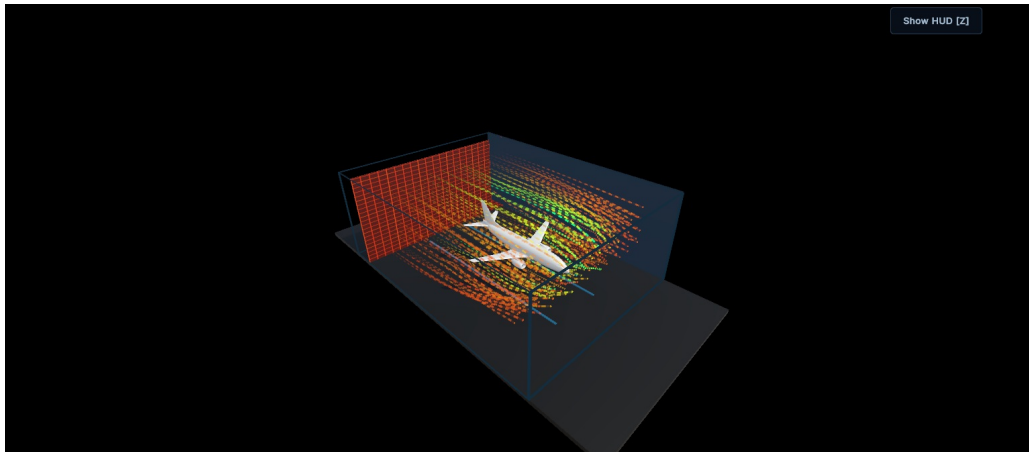
MODE-SPECIFIC RESULTS

Metric	Result
Regime	Turbulent
Assessment	Flow field not fully stable; solver divergence is elevated.
Rating	Excellent
Score	100%
Suggestions	Increase iterations/frame or lower turbulence intensity.



VISUALIZATION RESULTS

FLOW STREAMLINES



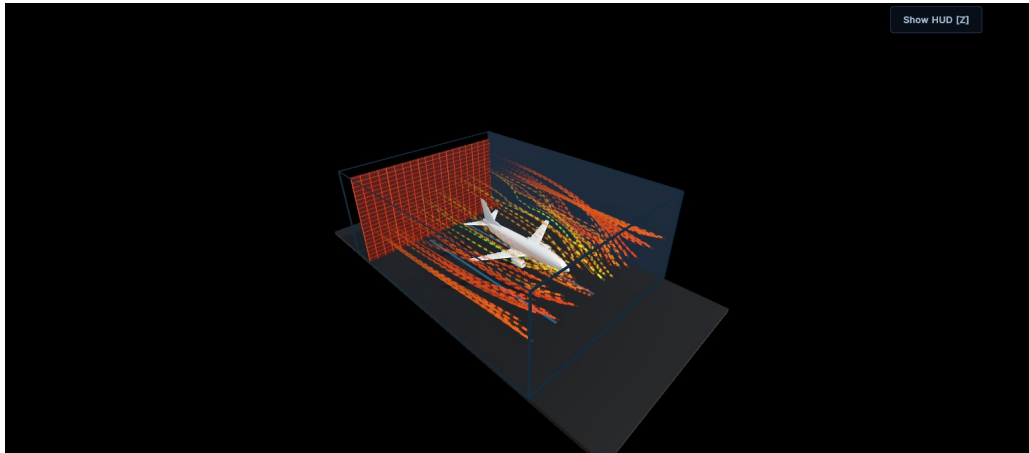
Primary streamline visualization of the active flow field.

Streamlines show the dominant path followed by the flow around the model and help reveal acceleration zones, wake shape, and large-scale separation behavior.

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HORIZONTAL STREAMLINES



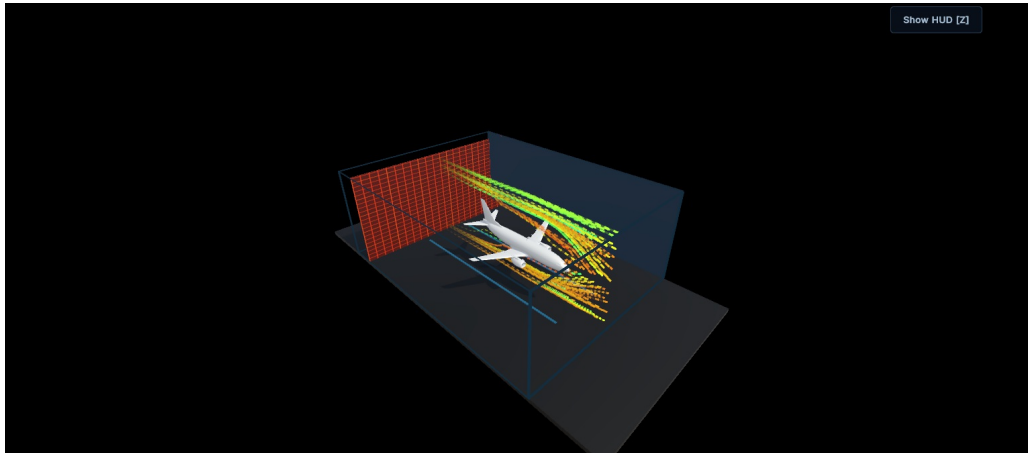
Horizontal streamline pass highlighting lateral flow behavior around the body.

This view emphasizes sidewash, lateral deflection, and horizontal wake development, making it easier to inspect symmetry and side-surface interaction.

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VERTICAL STREAMLINES



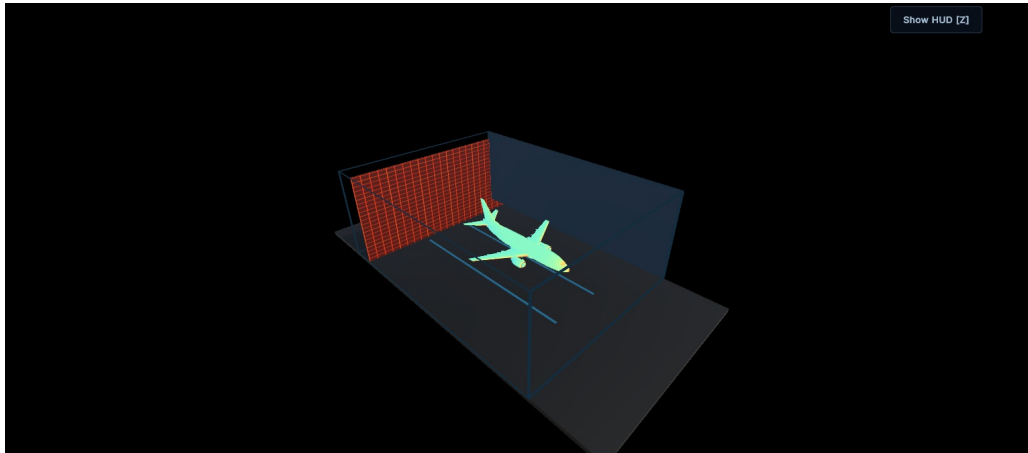
Vertical streamline pass highlighting roof flow, ground interaction, and wake rise.

This view is useful for identifying vertical deflection, roof acceleration, underbody interaction, and the way the wake lifts or settles behind the geometry.

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SURFACE PRESSURE



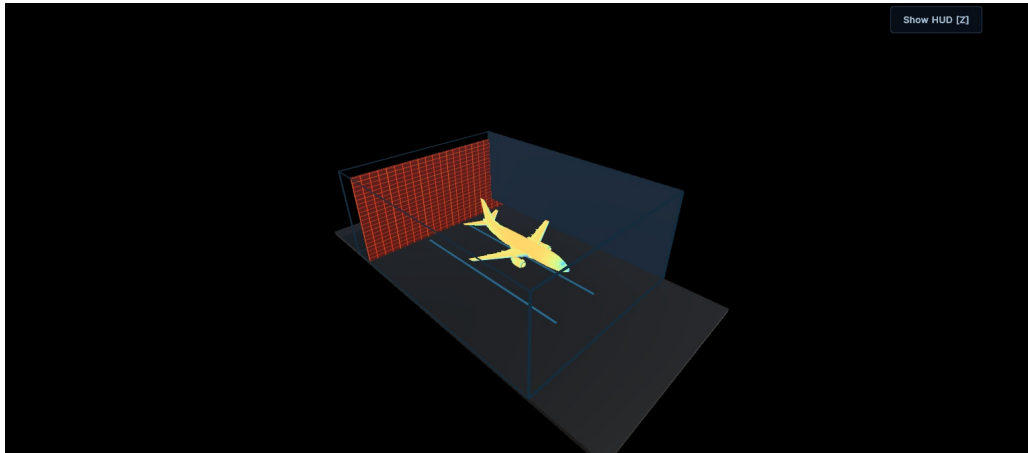
Surface pressure contour mapped directly onto the model body.

Surface pressure highlights stagnation zones, suction regions, and local loading trends that contribute directly to the pressure component of aerodynamic drag and lift.

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SURFACE FRICTION



Surface friction contour showing the wall-shear response on the body.

Surface friction indicates where the flow is strongly attached, where shear intensifies, and where the body is likely experiencing stronger skin-friction effects.

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CONCLUSION

Wind Tunnel Assessment: Solver divergence detected; results may require further refinement. Low pressure drop indicates a highly optimized flow path.

ASSESSMENT SNAPSHOT

Metric	Result
Simulation Mode	WindTunnel
Timestamp	300.9 s
Flow Regime	Turbulent
Assessment	Flow field not fully stable; solver divergence is elevated.
Quality Rating	Excellent
Quality Score	100%
Suggestions	Increase iterations/frame or lower turbulence intensity.

MODEL QUALITY

Metric	Result
Overall Score	75 / 100
Grade	Good
Predicted Cd Range	0.266 - 0.360
Separation Risk	27%
Downforce Potential	88%
ML Efficiency Score	91%
Feature Breakdown	Aspect Ratio: 3.15 [Excellent] (impact: 1%) Slenderness: 1.16 [Fair] (impact: 2%) Blockage: 0.21 [Good] (impact: 19%) Volume Eff.: 0.04 [Poor] (impact: 9%) Smoothness: 0.09 [Excellent] (impact: 17%) Symmetry: 0.93 [Excellent] (impact: 16%) Clearance: 0.79 [Excellent] (impact: 22%) Rear Taper: 0.60 [Excellent] (impact: 13%)

Suggested Improvements 1. Low volume efficiency (lots of concavities). Filling concave regions reduces separated flow zones.

This report was automatically generated by the FluidWorks CFD Simulation Platform. The results provided are based on the simulation parameters defined in the setup section. For detailed engineering optimization based on these results, please consult with an aerodynamicist.

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